

Md Mushfiqur Rahaman

Ph.D in Industrial Engineering (Ongoing)
Department of Industrial and Management Systems Engineering
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SUMMARY OF QUALIFICATIONS

Industrial & Automation Engineer with 6+ years in FMCG manufacturing (Unilever, Nestlé) leading Industry 4.0, SCADA/PLC, and plant digitalization. Delivered OPC/SQL/Azure-based monitoring and reporting systems, SPC solutions, and factory network backbones. Currently pursuing a PhD in Industrial Engineering (Data Science) at WVU.

JOB EXPERIENCE

• Unilever Bangladesh Ltd.

Manager-Electrical and Automation

March 2020-July 2023

Chittagong, Bangladesh

– Roles and Responsibilities

- * Lead for implementation of Industry 4.0-Digital Factory
- * Electrical and Automation Lead for the factory
- * Country Level Technical Assistance on Automation and Instrumentation (TA3 - Subject Matter Expert)
- * Country Level Technical Assistance on Electrical Safety and System Design (TA3 - Subject Matter Expert)
- * Key responsible person for utility operations and cost optimization (Generators, Chillers, Compressors, WTP, and Boilers)
- * Departmental CAPEX and project co-ordinator

– Key Significant Projects

- * Implementation of Energy Monitoring and Reporting system using local data server
- * Design of Central Automation System (CAS) Network- A complete physical network backbone for Industry 4.0
- * Implementation of Digital Factory Operating System (DFOS) app for packing lines efficiency management using Open Platform Communication (OPC) server and Microsoft Azure Cloud
- * Implementation of Statistical Process Control App for Process Lines using SQL server and SQL Server Reporting Services (SSRS)
- * Web Based Utility Monitoring and Reporting System using SCADA and SQL server
- * PCC panel replacement for three packing lines and two process lines
- * Electrical Safety Projects for Kalurghat Site
- * Commissioned two European bottle-filling and packing lines in the absence of OEM during the COVID lockdown

– Special Achievement

- * Awarded 'Best Engineer' of the year 2021
- * Received the Supply Chain Director's Award in 2022 for 'Talent Development in Control and Instrumentation Across the Company'

• Nestle Bangladesh Ltd.

Officer-Electrical and Automation

December 2018 to February 2020

Gazipur, Bangladesh

– Roles and Responsibilities

- * Maintaining Electrical and Automation maintenance in Noodle Manufacturing Plant
- * Maintaining the availability of electrical spare parts for all machines in the Noodle Manufacturing Lines in the technical store
- * Implementing improvement Automation project in overall plant

– Key Significant Projects

- * FreeWeigh.net Server (Mettler Toledo) commissioning for weighing system for realtime data collection and analysis
- * WIPOTEC-OCS Checkweigher installation and commissioning for Noodle Manufacturing Line
- * Installation and Commissioning Vertical of Filling and Wrapping Machine of TOPAC
- * Automation part Oil Cooling System in Noodle Line

- **Unipolar Automation Technologies (System Integrator of Rockwell Automation)** *March 2017-December 2018*
Project Engineer Dhaka, Bangladesh
 - **Roles and Responsibilities**
 - * Designing Industrial Control and Instrumentation Project as per the client's requirement
 - * Preparing Bill of Materials (BOM) for the Industrial Control and Instrumentation projects
 - * Installation and commissioning of the Automation project
 - **Key Significant Projects**
 - * PLC Lab Setup & Training for Dept of Mechatronics and Industrial Engineering at Chittagong University of Engineering and Technology
 - * Resin manufacturing plant end to end automation at Berger Paints Bangladesh (Role: Project Lead and Programmer)
 - * 132KV/33KV substation SCADA design and implementation at KYCR Steel Mill (Role: Project Lead and Programmer)
 - * Substation ACB Interlock System, Nestle Sreepur Plant, Nestle Bangladesh (Role: Programmer and Electrical Circuit Designer)
 - * Pump Station Automation (Pilot Project) of Dhaka Water Supply and Sewerage Authority

EDUCATION

- **West Virginia University** *Aug 2028(Ongoing)*
Ph.D. in Industrial Engineering (Concentration: Data Science)
- **West Virginia University** *Aug 2025*
Master of Science (M.S.) in Applied Mathematics
- **Chittagong University of Engineering and Technology** *March 2017*
Bachelor of Science (B.Sc.) in Electrical and Electronic Engineering

TECHNICAL SKILLS AND INTERESTS

PLC, HMI: Allen-Bradley, SIEMENS, Schneider
SCADA: Allen-Bradley, SIEMENS
VFD and Servo: Allen-Bradley, SIEMENS, Schneider, Fuji
Industrial IoT: OPC Server (Kepware), Azure (Basic Knowledge), Ethernet, Modbus, PROFINET, Cisco Firewall (Basic Knowledge)
Programming Language: C, Python, MATLAB, Java
Database: SQL, Oracle
CAD Software: AutoCAD
Cloud: Kubernetes
Server (installation and use): Windows Server, Linux Server, VMware Server, Proxmox
Engineering Simulation: Proteus (ISIS and ARES), P-Spice8.0, Cadence EDA Tool
ERP Software: SAP
Embedded Device: Microcontroller (PIC), Robotics Environment
Edge Computing and IoT: Qualcomm RB3 Gen2, Raspberry Pi, Arduino

PUBLIC SPEAKING AND SEMINAR

- **Keynote Speaker, 'Seminar on Landing on Industry with Future-fit Skillset'** *December 8, 2022*
Organized by the Department of Electrical and Electronic Engineering, CUET, on the occasion of EEE Day
- **Keynote Speaker, 'Seminar on IoT in Industrial Automation-The Journey towards Industry 4.0'** *Dec, 2021*
Organized by the Robo Mechatronics Association, CUET
- **Guest Speaker, 'An Open Talk on Science and Technology'** *February 20, 2017*
Organized by former students of Chittagong University Laboratory School and College

STUDENT ORGANIZATION

- **Co-Chair (Vice President)** *Dec 2015-Dec 2016*
IEEE Student Branch, CUET Chittagong, Bangladesh
- **Organizing Secretary** *Oct 2015-Dec 2016*
Robo Mechatronics Association, Bangladesh Chittagong, Bangladesh
- **President** *Sep 2015-Dec 2016*
Greater Barisal Student Association, CUET Chittagong, Bangladesh